

Multi-Objective Optimization for Multi-UAV-Assisted Mobile Edge Computing

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Abstract—Recent developments in unmanned aerial vehicles (UAVs) and mobile edge computing (MEC) have provided users with flexible and resilient computing services. However, meeting the computation-intensive and delay-sensitive demands of users poses a significant challenge due to the limited resources of UAVs. To address this challenge, we consider a multi-UAV-assisted MEC system. Based on this system, we formulate a multi-objective optimization problem aiming at minimizing the total task completion delay, reducing the total UAV energy consumption, and maximizing the total number of offloaded tasks. Since the problem is a mixed-integer non-linear programming (MINLP) and NP-hard problem, we propose a joint task offloading, computation resource allocation, and UAV trajectory control (JTORATC) approach. The problem is split into three components to cope with the coupling of these decision variables, and then solved individually to obtain the corresponding decisions. Specifically, the sub-problem of task offloading is solved by using distributed splitting and threshold

rounding methods, the sub-problem of computation resource allocation is solved by adopting the Karush-Kuhn-Tucker (KKT) method, and the sub-problem of UAV trajectory control is solved by employing the successive convex approximation (SCA) method. Simulation results show that the proposed JTORATC has superior performance compared with the other benchmark methods.

Index Terms—Unmanned aerial vehicles (UAV), mobile edge computing, task offloading, computation resource allocation, UAV trajectory control.

I. INTRODUCTION

WITH the rapid development of 6G and the Internet of Things (IoT), the number of smart mobile devices has shown unprecedented growth, leading to the increasing proliferation of various innovative mobile applications [1]. Most of these applications, such as face recognition, automatic navigation, and image processing, require intensive computation resources and low latency [2], [3]. However, handling the computation-hungry and real-time data generated by these applications is challenging due to the limited resources of mobile devices [4]. In this context, mobile edge computing (MEC) has been regarded as a promising solution that allows mobile devices to offload the computation-intensive and delay-sensitive tasks to proximate edge servers [5], [6], thereby reducing the computation burden, execution delay, and energy consumption of devices [7], [8], [9]. However, conventional MEC network relies on terrestrial infrastructures which are inflexible to deploy due to the installation cost and environmental limitations.

To overcome the physical restrictions of traditional terrestrial MEC systems, unmanned aerial vehicles (UAVs)-assisted MEC is emerging to offer flexible and low-cost offloading services due to the high maneuverability, flexibility, fast deployment, and line-of-sight (LoS) links of UAVs [10], [11], [12], [13]. By offloading the computation tasks to the proximate UAVs, mobile users can flexibly enjoy cloud computing services anywhere and anytime [14], [15], [16], [17].

However, designing an efficient task offloading approach in the multi-UAV-assisted MEC systems still faces several challenges. *First*, terrestrial MEC servers are typically deployed at fixed locations with limited coverage, causing service interruptions. Moreover, the single UAV-assisted MEC is deployed with limited computing resources and battery capacity, potentially making it difficult to handle a large amount of data or maintain

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long-term operations. *Second*, from the perspective of problem formulation, a significant challenge is that most current research tends to concentrate on optimizing a single performance metric, such as delay or energy consumption, or a combination of these two metrics while overlooking the multi-objective optimization. This could further compromise the overall effectiveness and efficiency of the system. *Third*, the UAV-assisted MEC system is characterized by several realistic features, including stringent and diverse requirements of users, constrained resources of UAVs, and other dynamics characteristics. These features present challenges in the joint optimization to satisfy the requirements of the users and the resource constraints of the UAVs. *Finally*, in terms of algorithm design, theoretic methods such as game theory possibly face challenges due to high computational and spatial complexity. Moreover, traditional heuristic methods such as swarm intelligence algorithms may encounter issues with a long iteration process and susceptibility to local optima. Additionally, machine learning approaches such as DRL potentially struggle with difficult convergence and long training periods, especially in large-scale scenarios.

To address the abovementioned challenges, this work proposes a multi-objective optimization problem of task offloading, computation resource allocation, and UAV trajectory control for multi-UAV assisted MEC systems. The main contributions are summarized as follows:

- *Edge Computing Architecture*: We consider a multi-UAV-assisted MEC system. Specifically, with the aid of MEC, resources of a UAV network are fully utilized to achieve a more flexible and efficient decision-making process, thereby optimizing the system performance. On the one hand, the MEC capability provided by the edge UAVs effectively compensates for the lack of flexibility of fixed MEC servers, and the computing resources can be dynamically deployed and reconfigured via UAVs to respond to real-time needs of different users in different regions. On the other hand, multi-UAV-assisted MEC effectively alleviates the constraints of a single UAV in terms of computing resources and coverage range, thereby significantly expanding the responsiveness and applicability of system.
- *Multi-objective Optimization Problem Formulation*: First, considering the delay-sensitive requirement of users, the energy limitation of the UAVs, and the stringent experience quality of the users, we formulate a multi-objective optimization problem. Specifically, this problem aims to minimize the total task completion delay, reduce the total UAV energy consumption, and maximize the total number of offloaded tasks by optimizing the decisions of task offloading, computation resource allocation, and UAV trajectory control. Moreover, to capture the dynamic of the considered scenario, the problem models the channel characteristics (i.e., small-scale fading, path loss, and shadowing). Consequently, the formulated problem is able to capture the realistic characteristics of the delay sensitivity, energy constraints, user experience, and channel dynamic within our scenario.
- *Joint Optimization Algorithm Design*: Since the optimization problem is a mixed-integer non-linear programming

(MINLP) and NP-hard problem, we propose a joint task offloading, computation resource allocation and UAV trajectory control (JTORATC) approach. First, we transform the original multi-objective optimization problem into a single-objective optimization problem to facilitate trade-offs of different objectives by leveraging the linear weighting method. Moreover, to manage the complexity induced by coupled and discrete-continuous mixed optimization variables, we split the problem into three sub-problems. For the sub-problem of task offloading, distributed splitting and threshold rounding methods are constructed to obtain the optimal solution. For the sub-problem of computation resource allocation, we apply the Karush-Kuhn-Tucker (KKT) method to achieve the optimal solution. Besides, for the UAV trajectory control sub-problem, the successive convex approximation (SCA) method is employed to dynamically optimize UAVs trajectories based on real-time environmental conditions and task requirements.

- *Effectiveness and Efficiency Validation*: The performance of JTORATC is verified through theoretical analysis and simulation. Specifically, the proposed JTORATC approach is able to converge to a sub-optimal solution in polynomial time. Moreover, simulation results demonstrate that the proposed JTORATC achieves superior performance in terms of the objective function value, total task completion delay, and total UAV energy consumption compared with several benchmark schemes. We also find that the proposed algorithm exhibits better scalability in the considered scenario.

The rest of this study is organized as follows. Section II reviews the related work. Section III introduces the proposed system model and problem formulation. Section IV presents the JTORATC approach. Moreover, the simulation results are given in Section V. Finally, Section VI summarizes the paper.

II. RELATED WORK

In this section, we review the research work that are related to edge computing architecture, optimization objectives, solution spaces, and optimization approaches. Moreover, we summarize the differences between the related works and this work in Table I.

A. Edge Computing Architecture

In recent years, MEC has become an area of research interests. By offloading delay-sensitive, computation-intensive and energy-intensive tasks to MEC servers, quality of service (QoS) of mobile user applications and services can be significantly improved. For example, Zhang et al. [18] studied energy-efficient computation offloading (EECO) mechanisms for MEC in 5G heterogeneous networks, where a macro base station (MBS) is equipped with an MEC server to support compute-intensive applications with a large number of end devices. Furthermore, Mao et al. [19] considered an MEC system with energy harvesting devices and proposed an online task offloading algorithm that aims to seek renewable energy sources to power mobile devices.

TABLE I
COMPARISON BETWEEN RELATED WORKS AND THIS WORK

Reference	Edge Computing Architecture		Optimization Objectives			Solution Spaces			Optimization Approaches			
	UAV-assisted MEC	Multi-UAV-assisted MEC	Service Delay	Energy Consumption	Offloaded Tasks	Task Offloading	Resource Allocation	UAV Trajectory Control	Distributed Splitting	Threshold Rounding	SCA	KKT
[18]	×	×	×	✓	×	✓	×	×	×	×	×	×
[19]	×	×	✓	×	×	✓	×	×	×	×	×	×
[20]	×	×	✓	✓	×	✓	✓	×	×	×	×	×
[21]	✓	×	×	✓	×	✓	×	×	×	×	✓	×
[22]	✓	×	×	✓	×	✓	×	✓	×	×	✓	✓
[23]	✓	✓	✓	×	×	✓	✓	×	×	×	×	✓
[24]	✓	✓	✓	×	×	✓	✓	×	×	×	×	×
[12]	✓	✓	×	✓	×	✓	✓	×	×	✓	×	×
[25]	×	×	✓	×	×	✓	✓	×	×	×	×	✓
[26]	✓	×	✓	✓	×	✓	✓	×	×	×	✓	×
[27]	✓	×	×	✓	×	✓	✓	×	×	×	×	×
[28]	✓	✓	×	×	×	✓	×	✓	×	×	✓	×
[29]	✓	✓	×	✓	×	✓	×	✓	×	×	×	×
[30]	✓	✓	✓	✓	×	×	✓	✓	×	×	×	×
[31]	✓	✓	✓	✓	×	✓	×	×	×	×	×	×
[32]	✓	✓	✓	✓	×	✓	✓	×	×	×	×	×
[33]	✓	✓	×	✓	×	✓	✓	✓	×	×	×	×
This work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Besides, Kuang et al. [20] investigated the joint optimization problem of partial offloading scheduling and resource allocation for MEC systems with multiple independent tasks, aiming to minimize the weighted sum of execution delay and energy consumption while ensuring the transmission power constraints of the tasks. However, the abovementioned works rely on fixed edge infrastructure MEC servers, which may not be applicable to the scenarios with inadequate infrastructures.

Given the limitations of fixed MEC servers, there has been a growing research for UAV-assisted MEC systems to provide computing services flexibly for end devices and to handle dynamic tasks efficiently. For example, Li et al. [21] explored the UAV-assisted edge computing, aiming to maximize the energy efficiency, where the UAV acts as a small cloud in the sky, collecting and processing computing tasks that need to be offloaded by users. Moreover, the authors in [22] focused on deploying a UAV equipped with an MEC server to serve multiple terminal devices (TDs) of IoT in a finite period. However, these studies mainly concentrated on single UAV-assisted MEC systems, which may not be applicable to the real-world scenarios, especially for the large-scale networks with dense users.

To overcome the abovementioned challenges, more research has focused on multi-UAV-assisted MEC environments. For example, Tun et al. [23] investigated a multi-UAVs assisted cooperative MEC system integrated with an MEC-enabled terrestrial base station (BS) to minimize the total delay experienced by mobile users. Guo et al. [24] proposed a software defined network enhanced cooperative multiple UAV-enabled aerial computing system to address the challenges in remote areas. Moreover, Ei et al. [12] studied a multi-UAV assisted two-stage MEC system, where the UAVs provide the computing and relaying services to mobile devices to support the requirements of energy-constrained devices.

In the aforementioned studies, the UAVs are either stationary or follow pre-determined trajectories, which may lead to sub-optimal performances of the system, particularly in environments requiring high adaptability and responsiveness. On the one hand, these UAV-assisted MEC architectures lack flexibility

and adaptability in dynamic environments where user locations and communication qualities change rapidly. On the other hand, although some studies focus on energy-efficient mechanisms, the fixed UAV paths can lead to extra cost of energy consumption. To address these limitations, we study a multi-UAV-assisted MEC system, where the trajectories of UAVs are designed by considering the user demands, environment dynamics, and the energy efficiency of the UAVs.

B. Optimization Objectives

The formulation of a joint optimization problem is crucial to accurately capture the characteristics of the system and define the problem to be solved. For example, Liu et al. [25] proposed an approach for edge-cloud orchestration computing that aims to minimize the average latency of tasks. Moreover, Yu et al. [26] investigated an innovative UAV-assisted MEC system involving the interaction between IoT devices, UAVs and edge cloud (EC). Their optimization objective is to minimize the weighted sum of service latency and UAV energy consumption of all IoT devices. Furthermore, Chen et al. [27] studied the problem of minimizing the energy consumption of UAV equipped with MEC servers as a mobile base station to serve users.

However, the abovementioned works mainly consider the delay, energy consumption, or a combination of these two metrics, while neglecting to jointly take into account delay, energy consumption, and total number of offloaded tasks. In contrast, our work comprehensively addresses the three objectives of delay, energy consumption, and the number of offloaded tasks in the problem formulation. This holistic consideration may ensure the delay-sensitive requirements of tasks, while maintaining efficient energy management of UAVs and the experience quality of users.

C. Solution Spaces

Considering the strict requirements of users and the limited computation resources MEC system, existing studies explored to optimize specific performance metrics by determining the

decisions such as task offloading, computation resource allocation, and UAV trajectory control. For example, Ji et al. [28] focused on multi-UAV-assisted wireless networks, aiming to jointly optimize cache placement, UAV trajectory and transmission power within a finite period for multi-user content delivery. Moreover, Lee et al. [29] investigated how to control the UAV-MEC trajectory and offloading ratio so that the mobile devices can perform effective data processing when located far from ground-based stations. Besides, Park et al. [30] considered the deployment and optimization of MEC-UAVs for subterahertz communication in remote areas.

Previous studies primarily focused on optimizing one or two aspects of task offloading, resource allocation, and UAV trajectory control. However, these partial focuses may hinder the full exploitation of the advantages of UAV-assisted MEC and result in sub-optimal solutions. To address the limitations of these studies, we jointly optimize the task offloading, resource allocation, and UAV trajectory control for the multi-UAV-assisted MEC system.

D. Optimization Approaches

To solve the complex optimization problem for task offloading, resource allocation, and UAV trajectory control, many researchers are devoted to adopting efficient approaches. For example, Wu et al. [31] proposed a discrete Stackelberg game approach with multiple leaders and followers, aiming to capture the hierarchical feature and discrete optimization in the multi-coalition UAV-assisted MEC networks. Akter et al. [32] presented a messy genetic algorithm (mGA) approach to obtain task offloading and computation resource allocation. Moreover, Li et al. [33] designed an embedded multi-agent deep reinforcement learning (DRL) algorithm to enable UAVs to autonomously learn trajectory control and computation offloading decisions, adapting to dynamic requirements of users.

However, when the Stackelberg game approach is adopted, the computational complexity of the decision may increase rapidly with the number of players and the size of strategy space. Moreover, swarm intelligence algorithms typically require numerous iterations to approach a near-optimal solution, leading to high execution delays, particularly for high-dimensional solution spaces. Furthermore, while DRL is effective for training agents to make decisions, it demands extensive environmental interaction and a large volume of sample data, making it costly in resource-constrained scenario, especially in large-scale scenarios or complex dynamic environments. Therefore, these approaches may be potentially unsuitable for the context of our work. We aim to design a low-complexity and high-efficiency algorithm to adapt to the delay-sensitive, resource-constraint scenarios while ensuring the performances of system. Different from the existing studies, we propose an approach by employing the combination of distributed splitting, threshold rounding, SCA, and KKT. By integrating these four approaches, our scheme strikes a balance between low-complexity and high-efficiency, making it highly suitable for real-world applications that demand strict requirements on response speed and resource utilization efficiency. Furthermore, our approach reduces the

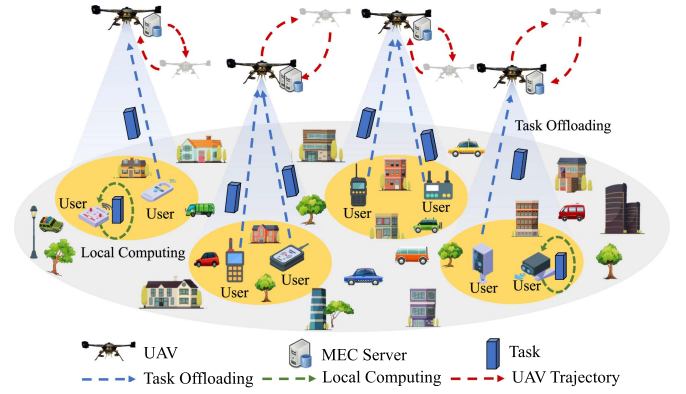


Fig. 1. Multi-UAV-assisted MEC for task offloading.

complexity associated with the traditional decision frameworks, such as existing algorithms and empirical models, thereby enhancing the adaptability and effectiveness of the system in dynamic environments.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we first present the system model, followed by the mobility model, communication model and computation model. Then, we formulate the multi-objective optimization problem. In addition, the notations associated with the system model are listed in Table II for the sake of readability.

A. System Model

As shown in Fig. 1, a multi-UAV assisted MEC system is considered, which consists of M rotary-wing UAVs and U users, denoted as $\mathcal{M} = \{1, 2, \dots, M\}$ and $\mathcal{U} = \{1, 2, \dots, U\}$. For ease of exposition, the continuous system timeline T is discretized into N time slots with equal time duration $\delta_t = T/N$, which is consistent with the coherence block of the wireless channel. Furthermore, each UAV is equipped with one or multiple MEC servers to provide computing services to the users. Besides, in each time slot, each user could generate a task that can be executed locally or offloaded to a UAV. The task of user u generated in time slot n is characterized as $\langle D_u[n], C_u, \tau_u \rangle$, wherein $D_u[n]$ is the task size (in bits), C_u is computation intensity of the task (cycles/bit), and τ_u denotes the deadline of the task.

B. Mobility Model

Without loss of generality, we consider a three-dimensional (3D) Cartesian coordinate system in which the horizontal position of each user $u \in \mathcal{U}$ is denoted as $\mathbf{w}_u = [x_u, y_u]^T$. Moreover, we consider that each UAV flies at a constant altitude H with the instantaneous horizontal position $\mathbf{q}_m = [x_m[n], y_m[n]]^T$, acceleration $\mathbf{a}_m[n]$, and velocity $\mathbf{v}_m[n]$ [34]. Let $V_{\max}$ be the maximum speed of the UAV, then the maximum distance that each UAV can travel between the two time slots is expressed as $D_{\max} = V_{\max} \delta_t$. In addition, it is critical to ensure a safe distance among UAVs to avoid collisions, which means that

TABLE II
SUMMARY OF NOTATIONS

Symbol	Description	Symbol	Description
U	The number of users	M	The number of UAVs
$\mathcal{U}$	The set of users	$\mathcal{M}$	The set of UAVs
H	UAV flight altitude	B	Channel bandwidth
$d_{u,m}[n]$	The distance between UAV m and user u in time slot n	$g_{u,m}[n]$	Channel power gain of UAV m and user u in time slot n
$R_{u,m}[n]$	The data transmission rate between user u and UAV m in time slot n	s	The rotor stiffness
p_u	Transmit power of user u	σ^2	Noise power
$D_u[n]$	The size of the task	C_u	The computation intensity of the task
f_u	The computing capability of user u	$f_{m,u}[n]$	The computing capability allocated by UAV m to user u
$T_{\text{local}}[n]$	The service delay of user u to offload its task to UAV m in time slot n	$T_{\text{off}}[n]$	The service delay of user u to process task locally in time slot n
$\tau_{m,u}^{\text{comp}}[n]$	Task processing delay	$\tau_{m,u}^{\text{trans}}[n]$	Task transmission delay
P_0	The blade profile power of UAV in hover state	P_{ind}	The induced power of UAV in hover state
U_{tip}	The blade tip speed of the rotor blade	v_0	The average rotor induced speed
d_0	The fuselage resistance ratio	A	The area swept by UAV blades
$E_m^{\text{fly}}[n]$	UAV m flight energy consumption	$E_{m,u}^{\text{comp}}[n]$	UAV m calculate energy consumption
V_{min}	The minimum speed of UAV m	V_{max}	The maximum speed of UAV m
κ_m	The effective switching capacitance of UAV m	κ_u	Effective switching capacitance of user u

the distance between any two UAVs should be greater than the minimum safe distance in any time slot. Therefore, the mobility constraints of UAVs can be given as follows:

$$\mathbf{q}_m[n+1] = \mathbf{q}_m[n] + v_m[n]\delta_t + \frac{1}{2}a_m[n]\delta_t^2, \quad \forall m, n, \quad (1a)$$

$$v_m[n+1] = v_m[n] + a_m[n]\delta_t, \quad \forall m, n, \quad (1b)$$

$$\|v_m[n]\| \leq V_{\text{max}}, \quad \|v_m[n]\| \geq V_{\text{min}}, \quad \forall m, n, \quad (1c)$$

$$\|\mathbf{q}_m[n] - \mathbf{q}_i[n]\| \geq D_{\text{min}}, \quad \forall m \neq i, n, \quad (1d)$$

$$\mathbf{q}_m[1] = \mathbf{q}_m[N], \quad \forall m, \quad (1e)$$

$$\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\| \leq D_{\text{max}} = V_{\text{max}}\delta_t, \quad \forall m, n, \quad (1f)$$

where (1a) and (1b) denote the position update and velocity update of UAVs, respectively, (1c) represents the speed constraints, (1d) ensures the safe distance between any two UAVs to avoid collisions, (1e) indicates that the initial and final positions of each UAV are fixed at the same location, and (1f) constrains the maximum flight distance during each time slot.

C. Communication Model

The channel power gain between a user and a UAV is calculated by incorporating the probabilistic LoS transmissions into both small-scale and large-scale fading [35]. For the uplink communication, the channel power gain between user u and UAV m at time slot n can be given as:

$$g_{u,m}[n] = p_{u,m}^{\text{LoS}} g_{u,m}^{\text{LoS}}[n] + (1 - p_{u,m}^{\text{LoS}}) g_{u,m}^{\text{NLoS}}[n], \quad (2)$$

where p^{LoS} represents the probability of LoS link, $g_{u,m}^{\text{LoS}}[n]$ and $g_{u,m}^{\text{NLoS}}[n]$ represent the channel power gain between user u and UAV m for LoS and NLoS links, respectively, which is calculated as:

$$g_{u,m}^{\text{LoS}}[n] = |h_{u,m}^{\text{LoS}}[n]|^2 (\mathcal{I}_{u,m}^{\text{LoS}}[n])^{-1} 10^{-\frac{\mathcal{J}_{\sigma}^{\text{LoS}}}{10}}, \quad (3a)$$

$$g_{u,m}^{\text{NLoS}}[n] = |h_{u,m}^{\text{NLoS}}[n]|^2 (\mathcal{I}_{u,m}^{\text{NLoS}}[n])^{-1} 10^{-\frac{\mathcal{J}_{\sigma}^{\text{NLoS}}}{10}}, \quad (3b)$$

where $h_{u,m}^{\text{LoS}}[n]$ and $h_{u,m}^{\text{NLoS}}[n]$, $\mathcal{I}_{u,m}^{\text{LoS}}[n]$ and $\mathcal{I}_{u,m}^{\text{NLoS}}[n]$, $\mathcal{J}_{\sigma}^{\text{LoS}}$ and $\mathcal{J}_{\sigma}^{\text{NLoS}}$ are the components of small-scale fading, path loss, and shadowing for LoS and NLoS links, respectively. For LoS link, these components are detailed as follows.

First, the small-scale fading characteristic of the channel is captured by using a parametric-scalable and good-fitting generalized fading model, i.e., Nakagami- m fading [36], and to distinguish it from the identification of UAV, we use w to represent the shape parameters in the Nakagami model, i.e., Nakagami- w . Specifically, $h_{u,m}^{\text{LoS}}[n]$ and $h_{u,m}^{\text{NLoS}}[n]$ follow the Nakagami distribution with fading parameter w^{LoS} and w^{NLoS} , which can be given as:

$$h_{u,m}^{\text{LoS}}[n] = \frac{2(w^{\text{LoS}})^{w^{\text{LoS}}} h_{u,m}^{\text{LoS}}[n]^{2w^{\text{LoS}}-1} e^{-\frac{w^{\text{LoS}} h_{u,m}^2}{\bar{p}}}}{\Gamma(w^{\text{LoS}}) \bar{p}^{w^{\text{LoS}}}}, \quad (4a)$$

$$h_{u,m}^{\text{NLoS}}[n] = \frac{2(w^{\text{NLoS}})^{w^{\text{NLoS}}} h_{u,m}^{\text{NLoS}}[n]^{2w^{\text{NLoS}}-1} e^{-\frac{w^{\text{NLoS}} h_{u,m}^2}{\bar{p}}}}{\Gamma(w^{\text{NLoS}}) \bar{p}^{w^{\text{NLoS}}}}, \quad (4b)$$

where $\bar{p}$ is the average received power in the fading envelope, and $\Gamma(w)$ is the Gamma function.

Second, the path loss between user u and UAV m for LoS/NLoS link can be given as:

$$\mathcal{I}_{u,m}^{\text{LoS}}[n] = \frac{(4\pi d_0 f_c)^2}{c^2} \left(\frac{d_{u,m}[n]}{d_0} \right)^{\beta^{\text{LoS}}}, \quad (5a)$$

$$\mathcal{I}_{u,m}^{\text{NLoS}}[n] = \frac{(4\pi d_0 f_c)^2}{c^2} \left(\frac{d_{u,m}[n]}{d_0} \right)^{\beta^{\text{NLoS}}}, \quad (5b)$$

where f_c is the carrier frequency, c is the speed of light, d_0 is the reference distance, $d_{u,m}[n]$ is the distance between user u and UAV m , and β^{LoS} and β^{NLoS} are the path loss exponents for LoS and NLoS links, respectively.

Finally, the shadowing captures the signal attenuation caused by shadowing in transmission, and it can be modeled as a zero-mean Gaussian distributed random variable, which is as follows:

$$\mathcal{J}_{\sigma}^{\text{LoS}}[n] \sim \mathcal{O}\left(0, (\sigma^{\text{LoS}})^2\right), \quad (6a)$$

$$\mathcal{J}_{\sigma}^{\text{NLoS}}[n] \sim \mathcal{O}\left(0, (\sigma^{\text{NLoS}})^2\right), \quad (6b)$$

where σ^{LoS} and σ^{NLoS} are the standard deviation of shadowing for LoS and NLoS links, respectively [35].

Accordingly, we employ the widely used orthogonal frequency-division multiple access (OFDMA) technology [37] to mitigate unreliable communication caused by interference and serve multiple users simultaneously. In time slot n , the data transmission rate between user u and UAV m can be given as:

$$R_{u,m}[n] = B \log_2 \left(1 + \frac{p_u g_{u,m}[n]}{\sigma^2} \right), \quad (7)$$

where B is the channel bandwidth, p_u is the transmit power of user u , and σ^2 is the noise power.

D. Computation Model

This section presents the service delay, energy consumption, and total number of offloaded tasks.

1) *Service Delay*: The service delay depends on the task offloading decision $a_{u,m}[n]$, which indicates whether the task of user u is offloaded to UAV m (i.e., $a_{u,m}[n] = 1$) or processed locally (i.e., $a_{u,m}[n] = 0$).

Local Computing: The service delay of user u to process task $K_u[n]$ locally in time slot n can be given as:

$$T_{\text{local}}[n] = \frac{D_u[n]C_u}{f_u}, \quad (8)$$

where f_u denotes the computing capability of user u .

UAV-Assisted Computing: The service delay of user u to offload task $K_u[n]$ to UAV m in time slot n mainly consists of the transmission delay and processing delay. Specifically, the transmission delay can be given as:

$$T_{u,m}^{\text{trans}}[n] = \frac{D_u[n]}{R_{u,m}[n]}. \quad (9)$$

Moreover, the processing delay can be expressed as:

$$T_{m,u}^{\text{comp}}[n] = \frac{D_u[n]C_u}{f_{m,u}[n]}, \quad (10)$$

where $f_{m,u}$ indicates the computing capability allocated by UAV m to the user u .

Therefore, by combining (9) and (10), the service delay for UAV-assisted computing can be calculated as:

$$T_{\text{off}}[n] = \frac{D_u[n]C_u}{f_{m,u}[n]} + \frac{D_u[n]}{R_{u,m}[n]}. \quad (11)$$

According to (8) and (11), the total task completion delay for all users across N time slots can be calculated as:

$$T_{\text{total}} = \sum_{n=1}^N \sum_{u=1}^U \sum_{m=1}^M a_{u,m}[n] T_{\text{off}}[n] + (1 - a_{u,m}[n]) T_{\text{local}}[n]$$

$$= \sum_{n=1}^N \sum_{u=1}^U \sum_{m=1}^M a_{u,m}[n] (T_{\text{off}}[n] - T_{\text{local}}[n]) + T_{\text{local}}[n]. \quad (12)$$

Given that the results are typically much smaller in comparison to the input data for most applications, the result download delay is ignored [38].

2) *Energy Consumption*: Completing tasks could impose additional costs of energy consumption on both users and UAVs.

Local Computing: The energy consumption of user u to process task $K_u[n]$ locally in time slot n can be given as:

$$E_{\text{local}}^{\text{comp}}[n] = \kappa_u (f_u)^2 D_u[n] C_u, \quad (13)$$

where κ_u denotes the effective capacitance coefficient of user u , which depends on the chip structure of the CPU [39].

UAV-Assisted Edge Computing: The energy consumed by UAV m to process task $K_u[n]$ in time slot n can be given as [40]:

$$E_{m,u}^{\text{comp}}[n] = \kappa_m (f_{m,u}[n])^2 D_u[n] C_u, \quad (14)$$

where κ_m represents the effective switching capacitance that depends on the CPU architecture.

UAV Flight: The propulsion power consumption of UAV m flying at speed $v_m[n]$ can be given as [41], [42]:

$$E_m^{\text{fly}}[n] = \left(P_0 \left(1 + \frac{3 \|v_m[n]\|^2}{U_{\text{tip}}^2} \right) + P_{\text{ind}} \left(\sqrt{1 + \frac{\|v_m[n]\|^4}{4v_0^4}} - \frac{\|v_m[n]\|^2}{2v_0^2} \right)^{1/2} + \frac{1}{2} d_0 \rho_0 s A \|v_m[n]\|^3 \right) \delta_t, \quad (15)$$

where P_0 is the blade profile power, P_{ind} is the induced power, U_{tip} is the blade tip speed of the rotor blade, and v_0 is the average rotor induced speed. Moreover, d_0 indicates the fuselage resistance ratio, ρ_0 is the air density, s is the rotor stiffness, and A indicates the area swept by UAV blades.

According to (14) and (15), the total energy consumption of the UAVs across N time slots is mainly incurred by task execution and flight, which can be given as:

$$E_{\text{total}} = \sum_{n=1}^N \left(\sum_{u=1}^U \sum_{m=1}^M (E_m^{\text{fly}}[n] + a_{u,m}[n] E_{m,u}^{\text{comp}}[n]) \right). \quad (16)$$

3) *Total Number of Offloaded Tasks*: The total number of offloaded tasks can reflect the service quality provided by UAVs, which can be calculated as:

$$K_{\text{total}} = \sum_{n=1}^N \left(\sum_{u=1}^U \sum_{m=1}^M a_{u,m}[n] D_u[n] \right). \quad (17)$$

E. Problem Formulation

This work aims to jointly minimize the total task completion delay, minimize the total UAV energy consumption, and maximize the total number of offloaded tasks by optimizing the decisions of task offloading $\mathbf{A} = \{a_{u,m}[n], \forall u, m, n\}$, computation resource allocation $\mathbf{F} = \{f_{m,u}[n], \forall u \in \mathbf{U}_0, m, n\}$, where $\mathbf{U}_0$ denote the set of users who select to offload tasks to UAV, and

UAV trajectory control $\mathbf{Q} = \{\mathbf{q}_m[n], \forall m, n\}$. Accordingly, the multi-objective optimization problem can be formulated as:

$$\mathbf{P} : \min_{\mathbf{A}, \mathbf{F}, \mathbf{Q}} \{T_{\text{total}}, E_{\text{total}}, -K_{\text{total}}\} \quad (18a)$$

$$\text{s.t. } a_{u,m}[n] \in \{0, 1\}, \forall u, m, n, \quad (18b)$$

$$\sum_{m=1}^M a_{u,m}[n] \leq 1, \forall u, n, \quad (18c)$$

$$\sum_{u=1}^U a_{u,m}[n] \leq 1, \forall m, n, \quad (18d)$$

$$a_{u,m}[n] \cdot T_{\text{local}}[n] \leq \tau_u, \forall u, m, n, \quad (18e)$$

$$a_{u,m}[n] \cdot T_{\text{off}}[n] \leq \tau_u, \forall u, m, n, \quad (18f)$$

$$0 \leq f_{m,u}[n] \leq f_m^{\max}, \forall m, n, \quad (18g)$$

$$\sum_{u=1}^{U_0} f_{m,u}[n] \leq f_m^{\max}, \forall m, n, \quad (18h)$$

$$(1a) - (1f). \quad (18i)$$

Constraints (18b)–(18d) represent the task offloading constraints of users. Constraints (18e) and (18f) indicate the deadline of the tasks. Constraints (18g) and (18h) limit the computation resource of UAV. Moreover, constraint (18i) represents the mobility constraint of UAV.

Proposition 1: Problem $\mathbf{P}$ is a non-convex and NP-hard MINLP.

Proof: Problem $\mathbf{P}$ involves binary variables (i.e., task offloading $\mathbf{A}$) and continuous variables (i.e., computation resource allocation $\mathbf{F}$ and UAV trajectory control $\mathbf{Q}$), while the inequalities in (1c) and (1d) are non-convex constraints. Consequently, problem $\mathbf{P}$ is an MINLP problem, which is also non-convex and NP-hard. ■

It can be deduced from Proposition 1 that it is difficult to find an optimal solution to problem $\mathbf{P}$ due to the NP hardness, which motivates us to propose the JTORATC.

IV. THE PROPOSED JTORATC

In this section, the JTORATC is proposed to solve the formulated optimization problem. First, we transform the original multi-objective optimization problem into a single-objective optimization problem to simplify complexity, unify evaluation criteria, and facilitate trade-offs of different objectives. Then, we split the single-objective optimization problem into three sub-problems of task offloading, computation resource allocation, and UAV trajectory control based on the block alternate descent method, which alternates the solution of each sub-problem iteratively. Specifically, we employ distributed splitting and threshold rounding methods to solve the sub-problem of task offloading. Moreover, the sub-problem of computation resource allocation is solved by adopting KKT method, and the sub-problem of UAV trajectory control is solved by employing SCA method.

A. Problem Reformulation

Solving problem $\mathbf{P}$ directly may presents several challenges. First, a multi-UAV assisted MEC scenario characterized by several realistic features, including the delay-sensitive requirements of users, limited resources of MEC servers, energy constraints of UAVs, and the dynamic of the system. These features complicate the design of a low complex and high efficient algorithm to ensure the stringent offloading requirements of users, meet the constrained computing resources of MEC servers, satisfy the limited energy of UAVs, and accommodate the dynamic characteristics of the system. Moreover, optimizing the metrics of $\mathbf{P}$ can often lead to conflicting outcomes. For example, reducing the task completion delay and increasing the number of offloaded tasks typically require higher allocations of computing resources, which in turn increases energy consumption. Conversely, typical approaches to decrease UAV energy consumption often involve reducing the computing resources allocated, resulting in increased delays and decreased number of offloaded tasks. To address these challenges, we leverage one of the multi-objective optimization methods, which is the linear weighting method [43], [44] due to its lower algorithmic complexity and its ability to yield effective results rapidly. Moreover, it also enables us to balance the trade-offs efficiently and achieve a satisfactory solution within the constraints of our problem. Consequently, the multi-objective optimization problem is transformed into a single-objective optimization problem as follows:

$$\begin{aligned} \rho(\mathbf{A}, \mathbf{F}, \mathbf{Q}) &= w_1 T_{\text{total}} + w_2 E_{\text{total}} - w_3 K_{\text{total}} \\ &= \sum_{n=1}^N \sum_{u=1}^U \sum_{m=1}^M w_1 (a_{u,m}[n] (T_{\text{off}}[n] - T_{\text{local}}[n]) + T_{\text{local}}[n]) \\ &\quad + w_2 (E_m^{\text{fly}}[n] + a_{u,m}[n] E_{m,u}^{\text{comp}}[n]) - w_3 (a_{u,m}[n] D_u[n]), \end{aligned} \quad (19)$$

where the non-negative parameters w_1 , w_2 , and w_3 denote the weight factors of the delay, energy consumption, and number of offloaded tasks, respectively. Note that the weight coefficients can be adjusted according to the user preferences on the objectives.

Remark 1: While Pareto frontier is able to address the multi-objective optimization problem by generating a set of non-dominated solutions across multiple optimization objectives, it is computationally complex [45] and may not be feasible for the delay-sensitive MEC system. This is particularly evident in large-scale scenarios with multiple decision objectives, where calculating and comparing all possible solution combinations can lead to significant delays and increased the energy consumption. Moreover, although the Pareto frontier approach provides better diversity and transferability [46], our proposed approach could be able to effectively and flexibly achieve a comparable effect in a significantly shorter time. In specific, by adjusting the weights according to the priorities on the optimization objectives and environmental features, our solutions can achieve diversity and adaptability, while also maintaining a lower computational overhead. Therefore, our approach could obtain the transferability and diversity at a lower cost, while reducing the complexity of the algorithm.

Based on (19), the original multi-objective optimization problem $\mathbf{P}$ can be transformed into a single-objective optimization problem $\mathbf{P}_1$, which is expressed as follows:

$$\mathbf{P}_1 : \min_{\mathbf{A}, \mathbf{F}, \mathbf{Q}} \rho(\mathbf{A}, \mathbf{F}, \mathbf{Q}) \quad (20a)$$

$$\text{s.t. } a_{u,m}[n] \in \{0, 1\}, \forall u, m, n, \quad (20b)$$

$$\sum_{m=1}^M a_{u,m}[n] \leq 1, \forall u, n, \quad (20c)$$

$$\sum_{u=1}^U a_{u,m}[n] \leq 1, \forall m, n, \quad (20d)$$

$$a_{u,m}[n] \cdot T_{\text{local}}[n] \leq \tau_u, \forall u, m, n, \quad (20e)$$

$$a_{u,m}[n] \cdot T_{\text{off}}[n] \leq \tau_u, \forall u, m, n, \quad (20f)$$

$$0 \leq f_{m,u}[n] \leq f_m^{\max}, \forall m, n, \quad (20g)$$

$$\sum_{u=1}^{U_0} f_{m,u}[n] \leq f_m^{\max}, \forall m, n, \quad (20h)$$

$$(1a) - (1f). \quad (20i)$$

In conclusion, this transformation can reduce the complexity of the original optimization problem, balance the trade-offs among different performance metrics, and achieve diversity among different objectives at a lower cost, which is more suitable for delay-sensitive and resource-constrained scenarios.

However, the transformed optimization problem is still difficult to solve due to the convex objective function and non-convex constraints. Considering that the decision variables of task offloading, computation resource allocation, and UAV trajectory control are coupled with each other, we split the single-objective optimization problem into three sub-problems of task offloading, computation resource allocation, and UAV trajectory control based on the block alternate descent method.

Remark 2: If problem $\mathbf{P}_1$ is split into three sub-problems, each sub-problem can be transformed into a convex optimization problem and can be solved using the existing methods. However, splitting problem $\mathbf{P}_1$ into two sub-problems may compromise flexibility, solving efficiency, and complexity. For example, if the original problem is split into the sub-problems of $\mathbf{P}_{1.1} : \min_{\mathbf{A}, \mathbf{F}} \rho(\mathbf{A}, \mathbf{F})$ and $\mathbf{P}_{1.2} : \min_{\mathbf{Q}} \rho(\mathbf{Q})$, or into the sub-problems of $\mathbf{P}_{1.1} : \min_{\mathbf{A}, \mathbf{Q}} \rho(\mathbf{A}, \mathbf{Q})$ and $\mathbf{P}_{1.2} : \min_{\mathbf{F}} \rho(\mathbf{F})$, combining binary variable $\mathbf{A}$ and continuous variable $\mathbf{F}$ or $\mathbf{Q}$ into a single sub-problem might limit the ability to apply different methods tailored to each variable type. Furthermore, if the problem is split into sub-problems of $\mathbf{P}_{1.1} : \min_{\mathbf{A}} \rho(\mathbf{A})$ and $\mathbf{P}_{1.2} : \min_{\mathbf{F}, \mathbf{Q}} \rho(\mathbf{F}, \mathbf{Q})$, the mixed continuous variables could further add the complexity on solving the problem. Thus, it is reasonable to split the original optimization problem into three sub-problems.

B. Task Offloading

Given the computation resource allocation $\hat{\mathbf{F}} = \{\hat{\mathbf{F}}_{m,u}[n], \forall u \in \mathbf{U}_0, m, n\}$ and UAV trajectory control $\hat{\mathbf{Q}} = \{\hat{\mathbf{Q}}_m[n],$

$\forall m, n\}$, the task offloading optimization problem can be expressed as:

$$\mathbf{P}_{1.1} : \min_{\mathbf{A}} \rho(\mathbf{A}, \hat{\mathbf{F}}, \hat{\mathbf{Q}}) \quad (21a)$$

$$\text{s.t. } a_{u,m}[n] \in \{0, 1\}, \forall u, m, n, \quad (21b)$$

$$\sum_{m=1}^M a_{u,m}[n] \leq 1, \forall u, n, \quad (21c)$$

$$\sum_{u=1}^U a_{u,m}[n] \leq 1, \forall m, n, \quad (21d)$$

$$a_{u,m}[n] \cdot T_{\text{local}}[n] \leq \tau_u, \forall u, m, n, \quad (21e)$$

$$a_{u,m}[n] \cdot T_{\text{off}}[n] \leq \tau_u, \forall u, m, n. \quad (21f)$$

Problem $\mathbf{P}_{1.1}$ is still non-convex due to the constraint (21b), which involves the binary variable $a_{u,m}[n]$. Therefore, we first transform problem $\mathbf{P}_{1.1}$ into a convex problem by using the distributed splitting method. Then, the threshold rounding method is adopted to convert the relaxed continuous variables back into binary variables.

For the non-convex constraint (21b), we first employ the distributed splitting method to transform the binary variables into continuous variables. In specific, the variable of task offloading $\mathbf{A}$ is split into two parts, i.e., $\mathbf{X}$ and $\mathbf{Y}$ as follows:

$$\mathbf{X} = \{x_a^u[n] \mid x_a^u[n] = 1 - a^u[n], \forall u, n\}, \quad (22a)$$

$$\mathbf{Y} = \{y_a^m[n] \mid y_a^m[n] = a^m[n], \forall m, n\}, \quad (22b)$$

where $a^u[n] = 1$ indicates that the task is processed locally, and $a^m[n] = 1$ means being offloaded to the UAV. Then, We slack the variables into continuous variables as follows:

$$\bar{\mathbf{X}} \triangleq \left\{ \sum_{u=1}^U x_a^u[n] = 1, x_a^u[n] \in [0, 1] \right\}, \quad (23a)$$

$$\bar{\mathbf{Y}} \triangleq \left\{ \sum_{m=1}^M y_a^m[n] = 1, y_a^m[n] \in [0, 1] \right\}. \quad (23b)$$

Based on the above steps, problem $\mathbf{P}_{1.1}$ is transformed into problem $\mathbf{P}_{1.1'}$ as follows:

$$\mathbf{P}_{1.1'} : \min_{\bar{\mathbf{X}}, \bar{\mathbf{Y}}} \rho(\bar{\mathbf{X}}, \bar{\mathbf{Y}}, \hat{\mathbf{F}}, \hat{\mathbf{Q}}) \quad (24a)$$

$$\text{s.t. } \sum_{m=1}^M y_a^m[n] \leq 1, \forall u, n, \quad (24b)$$

$$\sum_{u=1}^U x_a^u[n] \leq 1, \forall m, n, \quad (24c)$$

$$x_a^u[n] \cdot T_{\text{local}}[n] \leq \tau_u, \forall u, m, n, \quad (24d)$$

$$y_a^m[n] \cdot T_{\text{off}}[n] \leq \tau_u, \forall u, m, n, \quad (24e)$$

$$(23a), (23b). \quad (24f)$$

Proposition 2: Problem $\mathbf{P}_{1.1'}$ is a convex non-linear programming (NLP) problem.

Proof: Since the computation resource allocation and UAV trajectory control have been fixed, $E_m^{\text{fly}}[n]$ and $E_{m,u}^{\text{comp}}[n]$ are fixed values that do not affect the convexity of problem $\mathbf{P}_{1.1}'$. Consequently, the objective function ρ is the sum of three linear expressions about $x_a^u[n]$ and $y_a^m[n]$, rendering it a linear function. Besides, two variables (i.e., $\bar{\mathbf{X}}$ and $\bar{\mathbf{Y}}$) are involved in the sub-problem. Therefore, problem $\mathbf{P}_{1.1}'$ is a convex NLP problem [47]. ■

According to Proposition 2, solving problem $\mathbf{P}_{1.1}'$ directly remains challenging since it is a convex NLP. Therefore, we iteratively optimize $\bar{\mathbf{X}}$ and $\bar{\mathbf{Y}}$ by splitting $\mathbf{P}_{1.1}'$. Specifically, it can be expressed as:

$$\mathbf{P}_{1.1}^{\bar{\mathbf{X}}} : \min_{\bar{\mathbf{X}}} \rho(\bar{\mathbf{X}}, \hat{\mathbf{F}}, \hat{\mathbf{Q}}) \quad (25a)$$

$$\text{s.t.} \quad (24c), (24d), (23a). \quad (25b)$$

$$\mathbf{P}_{1.1}^{\bar{\mathbf{Y}}} : \min_{\bar{\mathbf{Y}}} \rho(\bar{\mathbf{Y}}, \hat{\mathbf{F}}, \hat{\mathbf{Q}}) \quad (26a)$$

$$\text{s.t.} \quad (24b), (24e), (23b). \quad (26b)$$

Then, the optimal solution of $\mathbf{X}^r$ can be obtained by applying CVX. However, $\mathbf{X}^r$ is a continuous variable within the closed interval of $[0,1]$ while the decision of task offloading is a binary variable. Hence, the threshold rounding method [48] is applied to transform the relaxed $\mathbf{X}^r$ into binary variables. Specifically, each element $x^* \in \mathbf{X}^r$ is transformed as:

$$x^* = \begin{cases} 1, & \text{if } x^* \geq \delta, \\ 0, & \text{otherwise,} \end{cases} \quad (27)$$

where $\delta \in (0, 1)$ is a positive rounding threshold.

Therefore, the optimal decisions of task offloading for users can be obtained as $\mathbf{X}^*$, and problem $\mathbf{P}_{1.1}^{\bar{\mathbf{X}}}$ is solved. Similarly, problem $\mathbf{P}_{1.1}^{\bar{\mathbf{Y}}}$ can be solved using the analogous method. As a result, the optimal solutions of task offloading can be obtained as $\mathbf{X}^*$ and $\mathbf{Y}^*$, and thus $\mathbf{A}^r$ can be obtained.

However, the integrity gap is significant because the rounding process from continuous variables to binary variables may violate the constraint. Therefore, to overcome this issue after rounding, we denote Δ_1 and Δ_2 as the maximum violation of constraints (24c) and (24d), respectively. As described in [49], the integrity gap ζ as follows:

$$\zeta = \min_{\bar{\mathbf{X}}} \frac{\rho}{\rho + \xi \Delta}, \quad (28)$$

where $\Delta = \Delta_1 + \Delta_2$, and ξ is the weight of Δ , i.e., the maximum violation of task offloading constraints. Besides, to ensure the stability and meaningfulness of the optimization problem, ξ is typically constrained to be non-negative, i.e., $\xi \geq 0$. Furthermore, the optimal solutions are accepted when $\zeta = 1$, which is proved by Theorem 1.

Theorem 1: There is no violation of the constraints when $\zeta = 1$.

Proof: Taking $\mathbf{X}^*$ as an example, constraints (24c) and (24d) are modified as follows:

$$\sum_{u=1}^U x_a^u[n] \leq 1 + \Delta_1, \quad (29a)$$

$$x_a^u[n] T_{\text{local}}[n] \leq \tau_u + \Delta_2, \quad (29b)$$

where Δ_1 and Δ_2 are obtained as follows:

$$\Delta_1 = \max \left\{ \sum_{u=1}^U x_a^u[n] - 1, 0 \right\}, \quad (30a)$$

$$\Delta_2 = \max \{ x_a^u[n] T_{\text{local}}[n] - \tau_u, 0 \}. \quad (30b)$$

We can derive from (28) that $\bar{\mathbf{X}}$ is compact, the solution of ρ is obtained through relaxing the variables $\bar{\mathbf{X}}$, and the solution of $\rho + \xi \Delta$ is obtained after rounding the relaxed variables. Then $\zeta \leq 1$ since 1 can be achieved by setting $\rho = \rho^*$. Therefore, the integrity gap can be bounded by $\zeta \leq 1$. Moreover, when ζ is closer to 1, it means that the selected rounded solution is at least as good as or better than an optimal solution. This indicates that we do not lost too much information or value due to rounding, and the obtained solution remains of high quality. As a result, the best rounding is achieved when ζ ($\zeta \leq 1$) is closer to 1 [12], [50]. In other words, $\zeta = 1$, when $\Delta_1 = 0$ and $\Delta_2 = 0$. Similarly, the same proof also applies to $\mathbf{Y}^*$. ■

C. Computation Resource Allocation

Given the optimized task offloading $\mathbf{A}^r = \{a_{u,m}^r[n], \forall u, m, n\}$ for users and the UAV trajectory control $\hat{\mathbf{Q}} = \{\hat{q}_m[n], \forall m, n\}$, the computation resource allocation problem can be expressed as:

$$\mathbf{P}_{1.2} : \min_{\mathbf{F}} \rho(\mathbf{A}^r, \mathbf{F}, \hat{\mathbf{Q}}) \quad (31a)$$

$$\text{s.t.} \quad 0 \leq f_{m,u}[n] \leq f_m^{\max}, \forall m, n, \quad (31b)$$

$$\sum_{u=1}^{U_0} f_{m,u}[n] \leq f_m^{\max}, \forall m, n. \quad (31c)$$

It can be deduced from Theorem 3 that (31a) is convex. The sub-problem of computation resource allocation is solved by adopting the KKT method. This is because KKT method can effectively deal with complex problems by transforming original problems into optimization problems with equality constraints, and improve the solving efficiency when considering multiple constraints.

Proposition 3: Problem $\mathbf{P}_{1.2}$ is a convex optimization problem.

Proof: The first-order derivative of ρ with respect to $f_{m,u}[n]$ is calculated as:

$$\frac{\partial \rho}{\partial f_{m,u}} = -\frac{a_{u,m}^r[n] D_u[n] C_u}{f_{m,u}[n]^2} + 2\kappa_m a_{u,m}^r[n] D_u[n] C_u f_{m,u}[n]. \quad (32)$$

The second-order derivative of ρ with respect to $f_{m,u}[n]$ is calculated as:

$$\frac{\partial^2 \rho}{\partial f_{m,u}^2} = -\frac{2a_{u,m}^r[n] D_u[n] C_u}{f_{m,u}[n]^3} + 2\kappa_m a_{u,m}^r[n] D_u[n] C_u. \quad (33)$$

It is clear that the second derivative $\frac{\partial^2 \rho}{\partial f_{m,u}^2} \geq 0$, and objective function ρ is a convex function with regard to $f_{m,u}[n]$. Therefore, problem $\mathbf{P}_{1.2}$ is convex and has an optimal solution. ■

Algorithm 1: Bisection Algorithm-Based Computation Resource Allocation.

Input: The maximum resource owned by the UAV $f_m^{\max}$.

Output: The optimal computation resource allocation $\mathbf{F}^r = \{f_{m,u}^r[n], m \in \mathcal{M}\}$.

```

1 Initialization: Search accuracy threshold:  $\varepsilon$ , the lower bound  $\lambda^{\min} = 0$  and the upper bound  $\lambda^{\max} = \lambda^{\text{bound}}$ ;
2 while  $\lambda^{\max} - \lambda^{\min} \geq \varepsilon$  do
3   Define  $\lambda = \frac{\lambda^{\min} + \lambda^{\max}}{2}$ ;
4   for  $u \in \mathbf{U}_0$  do
5     Compute  $f_{m,u}^r[n]$  by substituting  $\lambda$  into Eq. (35);
6   end
7   if  $f_{m,u}^r[n] \geq f_m^{\max}$  then
8      $\lambda^{\min} = \lambda$ ;
9   else
10     $\lambda^{\max} = \lambda$ ;
11  end
12 end
13 return  $\mathbf{F}^r = \{f_{m,u}^r[n], m \in \mathcal{M}\}$ .

```

Hence, the Slater condition is satisfied and the problem can be solved by using partial Lagrange function, which is formulated as:

$$L(F, \lambda) = \rho(\mathbf{A}^r, \mathbf{F}, \hat{\mathbf{Q}}) + \lambda \left(\sum_{u=1}^{\mathbf{U}_0} f_{m,u}[n] - f_m^{\max} \right), \quad (34)$$

where $\lambda \geq 0$ is the Lagrange multiplier related to the computation resource constraint of UAV.

Furthermore, the KKT conditions are used to obtain the optimal computation resource allocation $\mathbf{F}^r$. Specifically, by differentiating $L(F, \lambda)$ with respect to $f_{m,u}^r[n]$ and setting the result as 0, which can be expressed as:

$$\begin{aligned} 2\kappa_m a_{u,m}^r[n] D_u[n] C_u f_{m,u}^r[n]^3 + \lambda^* f_{m,u}^r[n]^2 \\ - a_{u,m}^r[n] D_u[n] C_u = 0. \end{aligned} \quad (35)$$

Then, the optimal computation resource allocation solution $\mathbf{F}^r$ can be achieved by applying the bisection algorithm [51], as shown in Algorithm 1. Specifically, the search accuracy threshold ε , lower bound $\lambda^{\min}$ and upper bound $\lambda^{\max}$ are set first (Line 1). Then, enter the loop of iterations until condition $\lambda^{\max} - \lambda^{\min} \geq \varepsilon$ is satisfied (Line 2). Furthermore, in each iteration, the algorithm determines the value of λ by bisection and performs the calculation of the computation resource allocation $f_{m,u}^r[n]$, then compares it with the maximum resource owned by the UAV $f_m^{\max}$ and updates λ if the condition is met (Lines 3 to 12).

D. UAV Trajectory Control

Given the optimized decisions of task offloading $\mathbf{A}^r = \{a_{u,m}^r[n], \forall u, m, n\}$ and computation resource allocation $\mathbf{F}^r = \{f_{m,u}^r[n], \forall u \in \mathbf{U}_0, m, n\}$, the UAV trajectory control problem

can be expressed as:

$$\mathbf{P}_{1.3} : \min_{\mathbf{Q}} \rho(\mathbf{A}^r, \mathbf{F}^r, \mathbf{Q}) \quad (36a)$$

$$\text{s.t. } \mathbf{q}_m[n+1] = \mathbf{q}_m[n] + v_m[n]\delta_t + \frac{1}{2}a_m[n]\delta_t^2, \forall m, n, \quad (36b)$$

$$v_m[n+1] = v_m[n] + a_m[n]\delta_t, \forall m, n, \quad (36c)$$

$$\|v_m[n]\| \leq V_{\max}, \quad \|v_m[n]\| \geq V_{\min}, \forall m, n, \quad (36d)$$

$$\|\mathbf{q}_m[n] - \mathbf{q}_i[n]\| \geq D_{\min}, \forall m \neq i, n, \quad (36e)$$

$$\mathbf{q}_m[1] = \mathbf{q}_m[N], \forall m, \quad (36f)$$

$$\|\mathbf{q}_m[n+1] - \mathbf{q}_m[n]\| \leq D_{\max} = V_{\max}\delta_t, \forall m, n. \quad (36g)$$

Since the constraints (36d) and (36e) are non-convex, the sub-problem $\mathbf{P}_{1.3}$ is non-convex. In view of the fact that the SCA method shows reliable convergence when dealing with non-convex optimization problems, and its property of effectively handling complex constraints, we choose to use SCA method to solve the sub-problem of UAV trajectory control, so as to improve the efficiency and speed of problem solving.

According to Lemma 1, we can transform the non-convex constraint into a lower bound on a convex function.

Lemma 1: For a given local point x^r , we have the following inequality:

$$x^2 \geq 2x^r(x - x^r) + (x^r)^2. \quad (37)$$

Proof: we first define a quadratic function as follows: $f(x) = x^2$. It can be easily observed that $f(x)$ is a convex function. Since any convex function can obtain its lower bound by adopting its first-order Taylor expansion at a local point, the inequality holds in (37) [28]. ■

For the non-convex constraint (36d), we first rewrite the non-convex constraints in (36d) as:

$$\|v_m[n]\|^2 \geq V_{\min}^2, \quad \|v_m[n]\|^2 \leq V_{\max}^2. \quad (38)$$

Furthermore, according to Lemma 1, given the local point $v_m^r[n]$, we have:

$$\begin{aligned} \|v_m[n]\|^2 &\geq \|v_m^r[n]\|^2 + 2(v_m^r[n])^T (v_m[n] - v_m^r[n]) \\ &\triangleq \vartheta_m^r[n]. \end{aligned} \quad (39)$$

Therefore, the non-convex constraint (36d) can be bounded as:

$$\vartheta_m^r[n] \geq V_{\min}^2, \quad \vartheta_m^r[n] \leq V_{\max}^2. \quad (40)$$

For the non-convex constraint (36e), we introduce an auxiliary variable $D_{m,i}[n]$ as:

$$D_{m,i}[n] = \|\mathbf{q}_m[n] - \mathbf{q}_i[n]\|, \quad \forall n, m \neq i. \quad (41)$$

Furthermore, since the distance between any two UAVs in the system should be greater than or equal to the defined safe distance, (41) can be further rewritten as $\|D_{m,i}\|^2 \geq D_{\min}^2$. Thus, given the fixed point $D_{m,i}^r[n]$, we have:

$$\|D_{m,i}[n]\|^2 \geq 2(D_{m,i}^r[n])^T (D_{m,i}[n] - D_{m,i}^r[n])$$

$$+ \|D_{m,i}^r[n]\|^2. \quad (42)$$

Therefore, for the given points of UAV $\mathbf{q}_i^r[n]$ and $\mathbf{q}_m^r[n]$, the non-convex constraint (36e) can be converted to:

$$\begin{aligned} D_{\min}^2 &\leq \|\mathbf{q}_m^r[n] + \mathbf{q}_i^r[n]\|^2 + 2(\mathbf{q}_m^r[n] - \mathbf{q}_i^r[n])^T \\ &\quad \times (\mathbf{q}_m[n] - \mathbf{q}_i[n] - \mathbf{q}_m^r[n] + \mathbf{q}_i^r[n]) \\ &\triangleq \zeta_{m,i}^r[n], \quad \forall n, m \neq i. \end{aligned} \quad (43)$$

In conclusion, according to (40) and (43), sub-problem $\mathbf{P}_{1.3}$ can be transformed into a convex problem $\mathbf{P}_{1.3'}$ by introducing the auxiliary variables $\vartheta_m^r[n]$ and $\zeta_{m,i}^r[n]$, which is as follows:

$$\mathbf{P}_{1.3'} : \min_{\mathbf{Q}} \rho(\mathbf{A}^r, \mathbf{F}^r, \mathbf{Q}) \quad (44a)$$

$$\text{s.t. } \vartheta_m^r[n] \geq V_{\min}^2, \vartheta_m^r[n] \leq V_{\max}^2, \forall m, n, \quad (44b)$$

$$\zeta_{m,i}^r[n] \geq D_{\min}^2, \forall n, m \neq i, \quad (44c)$$

$$(36b), (36c), (36f), (36g). \quad (44d)$$

It can be directly solved by existing optimization software such as CVX [52].

Theorem 2: Problem $\mathbf{P}_{1.3'}$ does not change the optimality of problem $\mathbf{P}_{1.3}$, i.e.,

$$\rho_{1.3'}(\mathbf{A}^r, \mathbf{F}^r, \mathbf{Q}) \leq \rho_{1.3}(\mathbf{A}^r, \mathbf{F}^r, \mathbf{Q})$$

Proof: By applying the lower bound results in (40) and (43), it can be deduced that the set of feasible solutions for subproblem $\mathbf{P}_{1.3'}$ is included within that for $\mathbf{P}_{1.3}$. Accordingly, the objective function value obtained from $\mathbf{P}_{1.3'}$ is always less than or equal to that from $\mathbf{P}_{1.3}$, which indicates that there exists at least one local optimal solution [28]. ■

E. Main Steps of JTORATC and Analysis

In this section, the main steps of the JTORATC are shown in Algorithm 2. First, the decision parameters of task offloading, computation resource allocation, UAV trajectory, and iteration numbers are initialized (Line 1). Then, the optimal decisions of task offloading, computation resource allocation and UAV trajectory control are obtained by solving the sub-problems $\mathbf{P}_{1.1}$, $\mathbf{P}_{1.2}$ and $\mathbf{P}_{1.3}$ alternately (Lines 2 to 8). In addition, the convergence and computational complexity of JTORATC are proved respectively.

1) *Convergence and Computation Complexity:* The convergence of JTORATC is analyzed as follows:

$$\rho(\mathbf{A}^r, \mathbf{F}^r, \mathbf{Q}^r) \stackrel{(a)}{\geq} \rho(\mathbf{A}^{r+1}, \mathbf{F}^r, \mathbf{Q}^r), \quad (45a)$$

$$\stackrel{(b)}{\geq} \rho(\mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \mathbf{Q}^r), \quad (45b)$$

$$\stackrel{(c)}{\geq} \rho(\mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \mathbf{Q}^{r+1}). \quad (45c)$$

Since task offloading is solved for a given resource allocation and UAV trajectory, the inequality (45a) holds. Furthermore, inequality (45b) is satisfied because the KKT condition is introduced to solve resource allocation sub-problem. Moreover, the

Algorithm 2: JTORATC.

Input: Task offloading $\mathbf{A}^0$, computation resource allocation $\mathbf{F}^0$, UAV trajectory $\mathbf{Q}^0$, and iteration index r^0 .

Output: Optimal task offloading $\mathbf{A}^{opt}$, computation resource allocation $\mathbf{F}^{opt}$ and UAV trajectory $\mathbf{Q}^{opt}$.

- 1 **Initialization:** The iteration index $r = 1$, the tolerance of accuracy $\epsilon = 10^{-4}$, $\mathbf{A}^0 = \emptyset$;
 - 2 **repeat**
 - 3 Fix $\{\mathbf{F}^r, \mathbf{Q}^r\}$, obtain the optimal solution of the problem $\mathbf{P}_{1.1}$ is $\mathbf{A}^{r+1}$;
 - 4 Fix $\{\mathbf{A}^{r+1}, \mathbf{F}^r, \mathbf{Q}^r\}$, obtain the optimal solution of the problem $\mathbf{P}_{1.2}$ based on Algorithm 1 is $\mathbf{F}^{r+1}$;
 - 5 Fix $\{\mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \mathbf{Q}^r\}$, obtain the optimal solution of the problem $\mathbf{P}_{1.3}$ is $\mathbf{Q}^{r+1}$;
 - 6 Update $\{\mathbf{A}, \mathbf{F}, \mathbf{Q}\}^r \leftarrow \{\mathbf{A}^{r+1}, \mathbf{F}^{r+1}, \mathbf{Q}^{r+1}\}$ and $r = r + 1$;
 - 7 **until** $\frac{\rho_{r+1} - \rho_r}{\rho_r} \leq \epsilon$;
 - 8 **return** $\{\mathbf{A}^{opt}, \mathbf{F}^{opt}, \mathbf{Q}^{opt}\}$.
-

inequality (45c) is due to the sub-optimality of UAV trajectory. Besides, the objective function $\rho(\mathbf{A}, \mathbf{F}, \mathbf{Q})$ is always positive owing to its non-negative expression. Therefore, the objective function is always non-increasing after every iteration, which is also finitely lower-bounded.

The computational complexity of problem $\mathbf{P}_{1.1}$ is $\mathcal{O}(K(\log_2(\frac{1}{\epsilon} + 1)))$ [53] based on the analysis of Theorems 2 and 1, where $K = U + M$, U is the number of users, M is the number of UAVs, and ϵ is the tolerance of accuracy. Furthermore, problem $\mathbf{P}_{1.2}$ is solved based on the bisection algorithm presented in Algorithm 1, and its computational complexity can be calculated as $\mathcal{O}(\log_2(\frac{\lambda^{\max} - \lambda^{\min}}{\epsilon}))$ [51], where $\lambda^{\min}$ and $\lambda^{\max}$ are the lower and upper bounds of λ , respectively, and ϵ denotes the search accuracy. Moreover, problem $\mathbf{P}_{1.3}$ involves solving non-convex problem, and the computational complexity is $\mathcal{O}(M^{3.5} \log_2(\frac{1}{\epsilon}))$ [38]. Therefore, the proposed JTORATC approach converges in polynomial, and the computational complexity is $\mathcal{O}(I_c(K(\log_2(\frac{1}{\epsilon} + 1) + \log_2(\frac{\lambda^{\max} - \lambda^{\min}}{\epsilon})) + M^{3.5} \log_2(\frac{1}{\epsilon})))$, where I_c is the number of outer iterations of Algorithm 2.

Remark 3: Note that the complexity reduction for solving problem $\mathbf{P}_{1.1}$ comes with the performance degradation due to the narrower solution space resulting from problem dividing. This is realistic because complexity reduction often comes with performance degradation, as demonstrated in [54]. However, the proposed JTORATC still achieves a feasible solution that meets the stringent requirements of users while meeting the constraints of the system, and the reasons are as follows. First, although dividing the original problem into sub-problems reduces the solution space, the optimization objective and constraints remain unchanged for each sub-problem, ensuring the feasibility of the solution to satisfy the constraints of the original problem. Similar approaches have been adopted in the studies [55], [56], [57] to

TABLE III
SIMULATION PARAMETERS

Symbol	Meaning	Default value
f_u	Computing capability of user u	340 MHz
p_u	Transmission power of user u	30 dBm
τ_u	Deadline of the task	[0.1, 75] s
H	Fixed flight altitude of UAV m	100 m
$\mathbf{q}_1$	Initial position of UAV1	[800, 1200]
$\mathbf{q}_2$	Initial position of UAV2	[2000, 1000]
V_{min}	Minimum speed of UAV m	20 m/s
V_{max}	Maximum speed of UAV m	60 m/s
f_m^{max}	Computing capability of UAV m	1200 MHz
κ_m	Effective switching capacitance of UAV m	10^{-27}
D_{min}	Safe distance between UAVs	10 m
B	Channel bandwidth	20 MHz
σ^2	Noise power	-100 dBm
m^{LoS}/m^{NLoS}	Nakagami fading parameter for LoS/NLoS communication	2/1
c	Speed of light	3×10^8 m/s
f_c	Carrier frequency	5.9 GHz
d_0	Reference distance	1 m
β^{LoS}/β^{NLoS}	Path loss exponent for LoS/NLoS communication	2.42/4.28
$\sigma^{LoS}/\sigma^{NLoS}$	Standard deviation of shadowing for LoS/NLoS transmissions	3/4 dB

address the original problem. Moreover, the optimality of the solution for the sub-problems of task offloading, computation resource allocation, and UAV trajectory control is proved, indicating that the outcome of each sub-problem is optimal. Finally, the simulation results presented in the next section will also demonstrate the superiority of our algorithm in terms of the objective function value, total task completion delay, total UAV energy consumption, and total number of offloaded tasks.

V. SIMULATION RESULTS

In this section, simulation results are presented to validate the effectiveness of the proposed approach.

A. Simulation Setup

We perform simulations to verify the validity of our proposed JTORATC method. Specifically, all the simulations are performed in MATLAB R2022b on a desktop computer equipped with an Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz 2.21GHz and 8GB RAM.

Scenarios: We consider a multi-UAV-assisted MEC system where 2 UAVs, 4 UAVs and 6 UAVs are deployed to offer offloading services to 8 users in a 2500×3000 m² rectangular area. The timeline is set as $T = 100$ s which is divided into 50 time slots.

Parameters: The default values of the simulation parameters are listed in Table III.

Benchmarks: This work evaluates the proposed JTORATC in comparison with the following schemes.

- *Random offloading and JTORATC-based resource allocation and trajectory control (ROJRATC):* the task offloading strategies of users are decided randomly, while the computation resource allocation and the trajectories of UAVs are decided based on the proposed JTORATC.

- *Nearby offloading and JTORATC-based resource allocation and trajectory control (NOJRATC):* the tasks of each user are offloaded to the nearest UAV, while the computation resource allocation and the trajectory control are decided based on the proposed JTORATC.
- *Many-to-many matching-based offloading and JTORATC-based resource allocation and trajectory control (MOJRATC):* the task offloading strategies are decided by using a matching-to-matching mechanism [58], while the computation resource allocation and the trajectories of UAVs are decided based on the proposed JTORATC.
- *Equalizing resources, JTORATC-based offloading and trajectory control (ERJOTC):* the computation resource allocation of UAVs are decided averagely while the strategies of task offloading and trajectory control are determined based on the proposed JTORATC.
- *JTORATC-based offloading and resource allocation, circular trajectory (JORACT):* the task offloading strategies and the computation resource allocation are decided based on the proposed JTORATC, while the UAVs fly following the circular trajectories.
- *JTORATC-based offloading and resource allocation, pre-defined trajectory (JORAPT):* the task offloading strategies and the computation resource allocation are decided based on the proposed JTORATC, while the UAVs follow pre-defined trajectories.

B. Evaluation Results

In this section, we first present simulation results to evaluate the performance of our proposed JTORATC. Then, we compare the effects of different scenario setups on the performance of the proposed JTORATC and benchmark schemes.

1) *System Performance:* Fig. 2(a), (b), (c), and (d) show the comparison results of the objective function value, total task completion delay, total UAV energy consumption, and total number of offloaded tasks of seven approaches with the increasing of the number of UAVs. It can be observed from Fig. 2 that as the number of UAVs increases, the objective function value and total UAV energy consumption of all approaches except the NOJRATC approach increase. This is because that the increase in the number of UAVs may lead to complex task scheduling, fierce competition for computing resources and cross-conflict of trajectories. However, considering the spatio-temporal relationship of tasks, the NOJRATC simplifies task scheduling, effectively uses resources and reduces conflicts on flight trajectories. Furthermore, the total number of offloaded tasks remains constant for all approaches as the number of UAVs increases. The reason is that we consider a multi-UAV-assisted MEC system, where each user produces one task during the entire time period. This means that the number of tasks generated by the users are fixed. That is, regardless of how many UAVs are deployed, the total number of tasks that they can handle remains constant. Therefore, the improvement of the overall task processing capability of UAVs does not increase the number of tasks handled, but only affects the processing rate of tasks due to the increased computing resources.

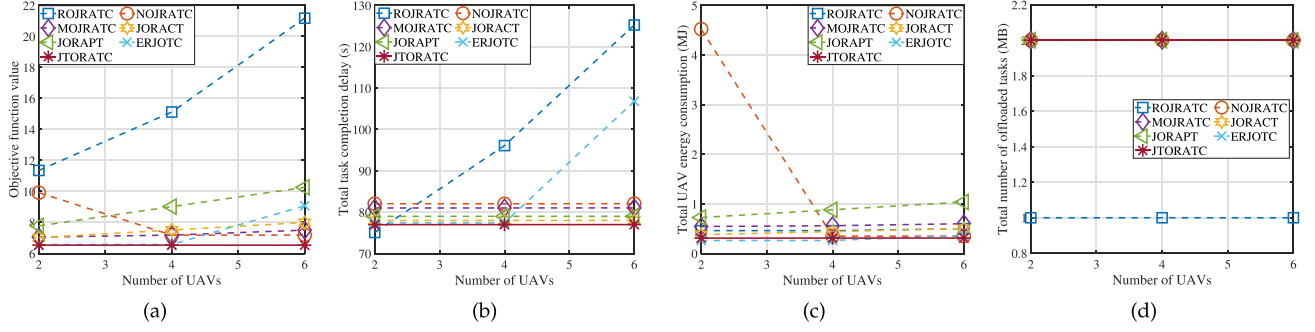


Fig. 2. System performance with UAV computation capacity. (a) Objective function value. (b) Total task completion delay. (c) Total UAV energy consumption. (d) Total number of offloaded tasks.

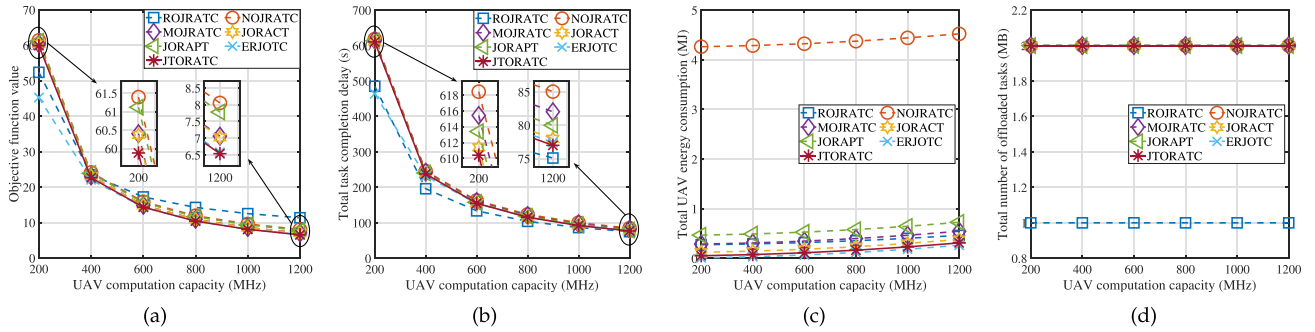


Fig. 3. System performance with numbers of UAVs. (a) Objective function value. (b) Total task completion delay. (c) Total UAV energy consumption. (d) Total number of offloaded tasks.

Moreover, the proposed JTORATC maintains superior performance compared to other approaches as the number of UAVs increases, and this can be attributed to several reasons as follows. First, ROJRATC, NOJRATC and MOJRATC mainly focus on optimizing the task offloading. The random offloading of ROJRATC, the offloading of the nearest MEC server of NOJRATC, and the matching offloading of MOJRATC may lead to inflexibly resource allocation of the MEC servers, triggering congestion and excessive resource usage. Second, ERJOTC mainly focuses on optimizing the computation resource allocation. It can be observed from Fig. 2(c) that as the number of UAVs increases, ERJOTC is slightly better than the proposed JTORATC in terms of total UAV energy consumption, but far worse than the proposed JTORATC in terms of the objective function value and total task completion delay. This is because the proposed JTORATC allows for more flexible task collaborative optimization, adaptation to different task requirements and dynamic changes, as well as latency reduction through parallel processing, thus improving the overall system performance, while ERJOTC can avoid excessive concentration of resources on a few UAVs. Furthermore, JORACT and JORAPT mainly focuses on the UAV trajectory control. The circular trajectory of JORACT and predefined trajectory of JORAPT have poor performance compared with the proposed JTORATC. This is because the proposed JTORATC allows dynamic adjustment of paths to accommodate real-time changes, resulting in more efficient and flexible task execution.

Accordingly, it can be concluded that the proposed JTORATC achieves optimal performance in multi-UAV-assisted MEC systems compared to other approaches.

2) *Performance Evaluation:* In this section, we evaluate the impact of different parameters on the system performance based on two UAVs. In this process, we keep the rest of parameters with default values and only vary the UAV computation capacity, task computation intensity, task size, and number of users. Through such exploration, we further verify that our proposed JTORATC approach has significant robustness and scalability in the system.

1) *Impact of UAV Computation Capacity:* Fig. 3(a), (b), and (c) show the effects of UAV computation capacity on the objective function value, total task completion delay, total UAV energy consumption, and total number of offloaded tasks. It can be observed from Fig. 3(a) that when the UAV computation capacity is less than 400 MHz, the ERJOTC approach achieves a lower objective function value than the proposed JTORATC. However, it is essential to recognize that while ERJOTC may find a local optimal solution in certain cases, ERJOTC cannot guarantee a global optimal solution. Therefore, this discrepancy is expected. In addition, as the computation capacity of UAVs increases, the proposed JTORATC is superior to other schemes in terms of objective function. The reason is that JTORATC jointly optimizes task offloading, computing resource allocation, and UAV trajectory control to adapt the computing capabilities of different UAV, resulting in satisfactory performance.

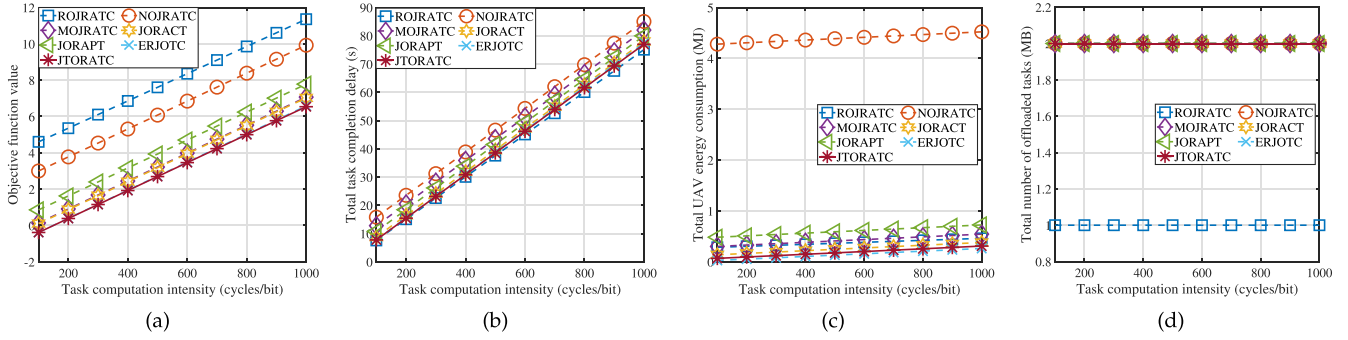


Fig. 4. System performance with task computation intensity. (a) Objective function value. (b) Total task completion delay. (c) Total UAV energy consumption. (d) Total number of offloaded tasks.

From Fig. 3(b) and (c), it is worth noting that when the UAV computation capacity is less than 600 MHz, the ROJRATC approach obtains a lower total task completion delay than the proposed JTORATC. Similarly, this outcome is expected since the proposed JTORATC is constrained by insufficient computing resources, leading to longer task completion times. Besides, with an increase in the UAV computation capacity, ROJRATC achieves a similar total completion delay to the proposed JTORATC by using the random offloading approach. However, the excessive focus on load balancing in ROJRATC causes higher energy consumption, leading to the relatively elevated energy consumption of UAVs.

From Fig. 3(d), we can see that for the total number of offloaded tasks, the seven approaches all show a constant trend regardless of the varying UAV computation capacity, since the total number of tasks completed by the UAV depends on the number of tasks generated by the users, and is independent from the computation capacity of UAV.

In summary, the results in Fig. 3 illustrate the adaptability of the proposed JTORATC with varying computing capacities of UAVs.

2) *Impact of Task Computation Intensity*: Fig. 4(a), (b), (c), and (d) reveal the effects of task computation intensity on the objective function value, total task completion delay, total UAV energy consumption, and total number of offloaded tasks. As can be seen, as the computational intensity increases, the seven approaches exhibit similar upward trends in terms of objective function value, total task completion delay and total UAV energy consumption. This is because higher computational intensity indicates heavier system workloads, which further leads to higher computational requirements. Consequently, more frequent task offloading, computation resource allocation and adjustments of UAV trajectory are necessary, which potentially results in additional overhead on computation and energy consumption.

Moreover, it can be seen from Fig. 4(a) that the proposed JTORATC outperforms other approaches in terms of objective function value. This is because JTORATC iteratively optimizes the decisions of task offloading, computation resource allocation, and UAV trajectory control to adapt to scenarios with varying workloads. From Fig. 4(b), we can observe that with the increasing of task computation intensity, ROJRATC achieves a similar total completion delay to the proposed JTORATC by using the random offloading approach. However, the

performance of ROJRATC in minimizing the objective function value and total UAV energy consumption is not as good as our proposed JTORATC approach. This is because the ROJRATC mainly relies on a random offloading strategy, which could not provide sufficient performance benefits in terms of the objective function, completion delay, and energy consumption, as the computational intensity of the tasks increases. In contrast, the proposed JTORATC takes a more comprehensive approach for system optimization.

From Fig. 4(c), it can be observed that ERJOTC shows optimal performance in terms of total UAV energy consumption. This is because ERJOTC helps to ensure the load balance of each UAV during task execution, reduce the excessive use of computing resources, and achieve an even distribution of overall computing energy consumption. However, the performance of ERJOTC is slightly inferior to our proposed JTORATC approach in objective function value and total task completion delay aspects. While from Fig. 4(d), we can see that for the total number of offloaded tasks, the seven approaches all show a constant trend regardless of the varying task computation intensity. Similarly, because the total number of tasks completed by the UAV depends on the number of tasks generated by the users, and has nothing to do with the task computation intensity.

In conclusion, the simulation results in Fig. 4 indicate the superiority of the proposed JTORATC in both light and heavy computational scenarios.

3) *Impact of Task Size*: Fig. 5(a), (b), (c), and (d) depict the effects of task size on the objective function value, the total task completion delay, the total UAV energy consumption, and the total number of offloaded tasks. It can be seen from Fig. 5 that the seven approaches exhibit upward trends as the task size increases. This is because the increasing task size indicates heavier workloads, which could further lead to higher costs of processing delay and energy consumption.

From Fig. 5(a), (b), and (c), it can be observed that when the task size is small (less than 1.5 MB), the performance of ERJOTC is better than the proposed JTORATC approach in objective function and total task completion delay aspects. However, when the task size is large (more than 3 MB), ERJOTC has the worst performance of all approaches. Moreover, for the total UAV energy consumption, ERJOTC shows the best performance. This is because when the task size is small, resources are allocated uniformly, then ERJOTC may be more

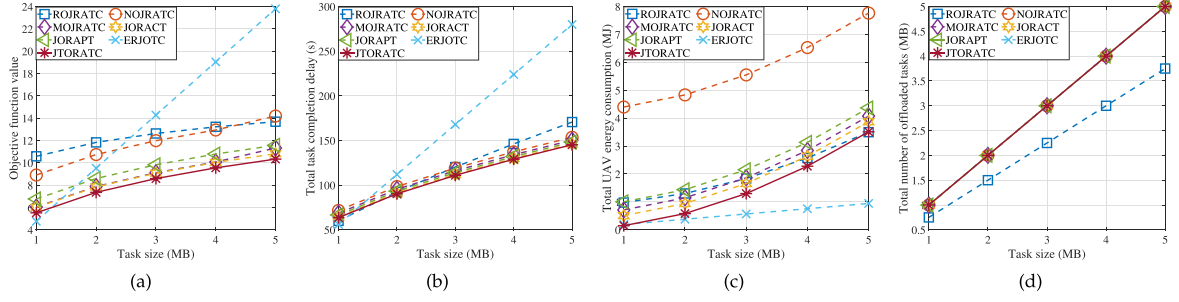


Fig. 5. System performance with task size. (a) Objective function value. (b) Total task completion delay. (c) Total UAV energy consumption. (d) Total number of offloaded tasks.

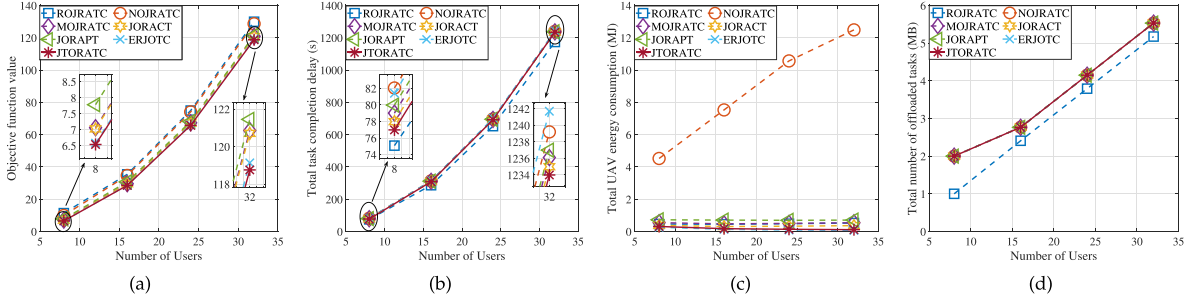


Fig. 6. System performance with number of users. (a) Objective function value. (b) Total task completion delay. (c) Total UAV energy consumption. (d) Total number of offloaded tasks.

efficient. However, when the task size is large, ERJOTC does not fully consider the complexity and different requirements of the task, which may lead to performance degradation. In addition, the average allocation strategy of ERJOTC ensures a relatively uniform distribution of tasks among UAVs and avoids excessive loads, thus reducing the energy consumption of UAVs.

Moreover, it can be seen from Fig. 5(d) that the seven approaches all show an upward trend as task size increases for the total number of offloaded tasks, because the total number of tasks generated by users is increasing, and the characteristics and requirements of the tasks may also change.

For the total number of offloaded tasks, the seven approaches all show an upward trend as task size increases since the total number of tasks generated by users is increasing, and the characteristics and requirements of the tasks may also change.

Accordingly, the simulation results in Fig. 5 demonstrate the superiority of the proposed JTORATC to adapt to heavy-loaded scenarios.

4) *Impact of Number of Users:* Fig. 6(a), (b), (c), and (d) present that with the increasing of the number of users, the objective function value, the total task completion delay, the total UAV energy consumption, and the total number of offloaded tasks of the seven approaches gradually increase. The reason is that as the number of users increases, the total number of tasks that need to be handled in the system also increases. This leads to a surge in resource competition and an increase in the load of communication networks.

Fig. 6(a) illustrates that our proposed JTORATC approach achieves the best performance in minimizing the value of the objective function. This can be attributed to the fact that our proposed JTORATC successfully balances the system

parameters by employing iterative optimization, and thus obtains a significant advantage in the overall performance.

From Fig. 6(b), (c), and (d), it can be observed that ROJRATC performs best in terms of total task completion delay. The random nature of ROJRATC can lead to tasks being assigned and completed more quickly. However, it should be noted that ROJRATC performs relatively poorly in terms of total UAV energy consumption and total number of offloaded tasks. This suggests that we must fully balance energy consumption and task completion efficiency while considering delay. In contrast, our proposed JTORATC approach takes this balance into account. Moreover, the ERJOTC has a good performance in reducing the total UAV energy consumption, mainly because evenly allocating resources avoids some resources being idle or overloaded. At the same time, because of the uniform allocation of resources, there is a certain sacrifice in the task completion delay, which is far less than our proposed JTORATC approach. This may be because task offloading is not flexible enough or resources cannot be optimized for the specific needs of the task.

Consequently, this set of simulation results highlights the importance of our proposed JTORATC approach in terms of scalability and robustness of the system.

VI. CONCLUSION

In this work, we study the multi-objective optimization of task offloading, computation resource allocation and UAV trajectory control in multi-UAV-assisted MEC systems. We first formulate the multi-objective optimization problem with the aim of minimizing the total task completion delay, reducing the total UAV energy consumption, and maximizing the total number

of offloaded tasks. To solve the problem, we transform the multi-objective optimization problem into a single-objective optimization problem and then propose the JTORATC to solve the transformed problem at a lower cost. Simulation results demonstrate that the proposed JTORATC has superior performance in terms of the total task completion delay and total UAV energy consumption. Specifically, JTORATC demonstrates better adaptability to the scenarios with varying computing capacities of UAVs, and it shows superior scalability in both light and heavy workload scenarios.

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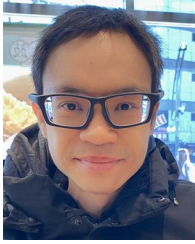
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